Where did the water go?

Learning intention

Use input minus collected water to estimate and compare water remaining after drainage.

We add the same amount.

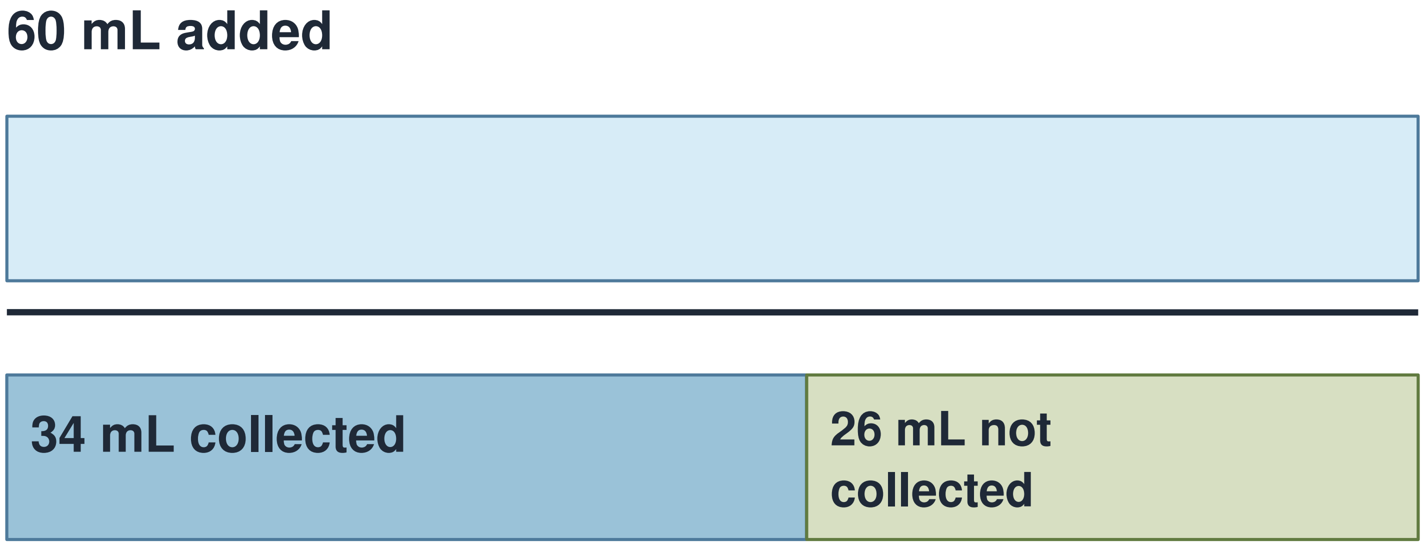
Different amounts drain out.

How much is not collected?

Three links from last time

1. Where can water move through soil?
2. Name one fair-test control.
3. What is 60 minus 20?

Account for the added water



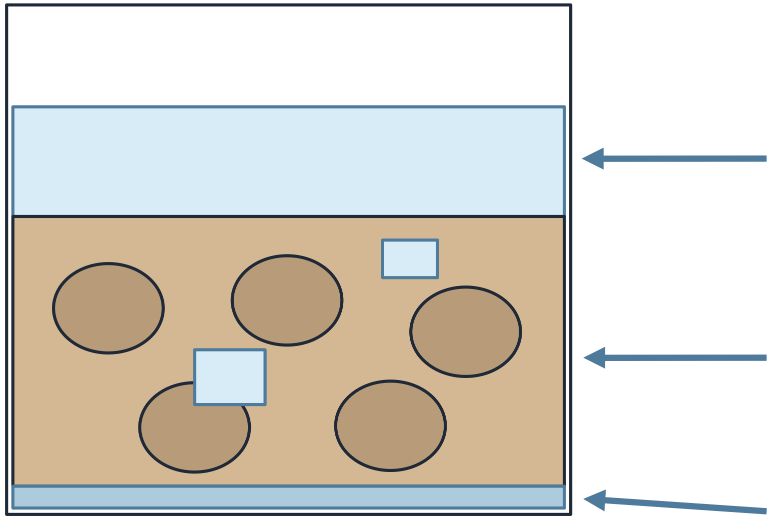
$$60 - 34 = 26 \text{ mL}$$

Model: initial water and losses ignored.

Start with the water added. Subtract the water collected. Record the difference in mL.

BUILD SCIENCE | I do | 2 min

Be precise about “held”



Check for standing water, drips, leaks and spills.

Water can remain in soil pores. Some may remain above the soil. The filter can hold water too.

Compare equal water budgets

Constructed sample data • 60 mL in each

Left at 5 min; initial water/losses ignored.

Each starts with 60 mL. Less collected means more left over. Always state the time. Which sample has the largest amount not collected?

The water-budget team

Read the inputs and outputs. Choose the correct calculation. Check with the whole-water bar.

Soil	Input (mL)	Collected at 5 min (mL)
A	60	46
B	60	34
C	60	22

Constructed sample data • no spills • initial water and evaporation ignored

For B: 60–34, 60+34 or 34–60?

Find the overclaim

“C held exactly 38 mL in the soil.” “C had 38 mL not collected at five minutes.” Which claim fits the evidence?

Check card	What happened?
P	No spills; same apparatus; no losses assumed
Q	A puddle remained above the soil
R	Some water escaped onto the tray

Constructed reasoning cards; consider each situation separately

Which statement is sound?

Each sample receives 60 mL. A collects 46 mL; C collects 22 mL.

A A has the greater amount not collected.

B C has the greater amount not collected at this time.

C C will therefore grow every plant best.

Investigate; record; explain

Choose paper data or a practical.

Choose calculator, writing or spreadsheet.

Use a number and a careful claim.

Keep the method matched

Same soil volume and preparation.

60 mL poured over 10 seconds.

Read at 5 minutes from pour start.

Make a water account

Input minus collected equals not collected.

Compare your three remainders.

Say what your estimate includes.

Check the claim before sharing

Have I used mL?

Have I stated 5 minutes?

Have I checked for losses?

Lundy loop: improve the claim

Share a result or a concern.

Listen for an overlooked location.

Rewrite one class conclusion.

What wording should our class keep?

Close the water account

E1. 60 mL in; 40 mL out. How much not collected?

E2. Where else could water remain?

E3. Why state the time?